

Pollinator YOLO Annotation Guidelines

1. Purpose and Scope

This document defines annotation rules for drawing bounding boxes around insects in Arctic field images in CVAT. The annotations are used to train a YOLO object detector for automated pollinator detection.

2. Annotation Tool

Annotations are created using a self-hosted CVAT server. Images are organized into tasks within the pollinator-yolo project. Each task contains a sequence of frames from a single camera folder, allowing frame-by-frame navigation to identify insects that may be hard to visualize in a single frame.

3. Annotation Classes

Label	When to use
bumblebee	Bumblebee (<i>Bombus</i>) visible
fly	Any fly (Diptera) visible
butterfly	Butterfly or moth visible
other	Any other insect visible (e.g. beetle)
unsure	Something moving could be an insect but undecidable even after checking surrounding frames; insect present but class is unclear and needs verification by a domain expert

Background frames containing no insects receive no bounding box.

4. Bounding Box Rules

- Draw one bounding box per insect. Do not rotate or tilt the box. The box should enclose the full insect body.
- If the legs are very long and would make the bounding box mostly background, draw the box around the body only and ignore most of the legs.
- If an insect is partially out of frame, still draw a bounding box around the visible portion.
- For very small or blurry insects, draw the box around the region where the insect is located. Label it unsure if the class cannot be determined.
- Use unsure only when necessary. When uncertain between a specific class and unsure, prefer the specific class.

5. Using Surrounding Frames

Each flagged frame comes with the three frames before and after it. Insects can be hard to spot in a single frame against the vegetation, but easy to see when flipping through several frames in a row, because the insect is the thing that moves.

Before labeling a difficult frame, flip through the surrounding frames to check whether the moving object is an insect and which class it belongs to. If something appears and disappears within a few frames, it is most likely a real insect.

6. Workflow

- Open a task in CVAT and navigate frame by frame using the arrow keys.
- For each frame, check whether an insect is visible.
- If no insect is visible, leave the frame without any bounding box and move on.
- If an insect is visible, draw a bounding box and assign the correct class.
- For ambiguous cases, scrub through surrounding frames before deciding on the class.
- Press Save regularly to avoid losing progress.
- For questions about specific cases, discuss in the group channel or consult the domain expert.
- Export format: Ultralytics YOLO Detection 1.0